

| Question           |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks Available |                     |          |          |          |          |
|--------------------|-----|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------------|----------|----------|----------|----------|
|                    |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2                 | AO3      | Total    | Maths    | Prac     |
| 7<br>FT<br>1<br>HT | (a) | (i)  |  | X marked where the 3 circles intersect                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                 | 1                   |          | 1        |          |          |
|                    |     | (ii) |  | Using <u>1</u> station means <u>many</u> possible epicentre locations / using <u>2</u> stations means there are <u>2</u> possible locations for the epicentre location (1)<br>Three stations are needed to eliminate the other options / three circles only cross at one place (1) Accept reference to triangulation                                                                                                                                                                                                                                                                          |                 | 2                   |          | 2        |          |          |
|                    | (b) |      |  | Station C (1)<br>as it has the smallest circle or diameter or radius / smallest distance or closest [to the epicentre] (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                 | 2                   |          | 2        |          |          |
|                    | (c) |      |  | The time delay for B is half that for A or converse (1)<br>Accept time is halved provided one arrival time is correct<br>[Time taken for P wave to arrive at station A = 10 s<br>Time taken for P wave to arrive at station B = 5 s so the time would be] <b>1:15:15</b> (1)<br>[The time lag would be half (25 s) the time lag for station A so the time S wave would arrive at station B] = <b>1:15:40</b><br>or $\left[\frac{60}{2} = 30\text{ s added to } 1:15:10\text{ to give}\right]$ <b>1:15:40</b> (1)<br><b>N.B. both calculations correct but no statement award 2 marks only</b> |                 | 1<br><br>1<br><br>1 | 1        | 3        | 2        |          |
|                    |     |      |  | <b>Question 7 FT</b><br><b>Question 1 HT total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>0</b>        | <b>7</b>            | <b>1</b> | <b>8</b> | <b>2</b> | <b>0</b> |